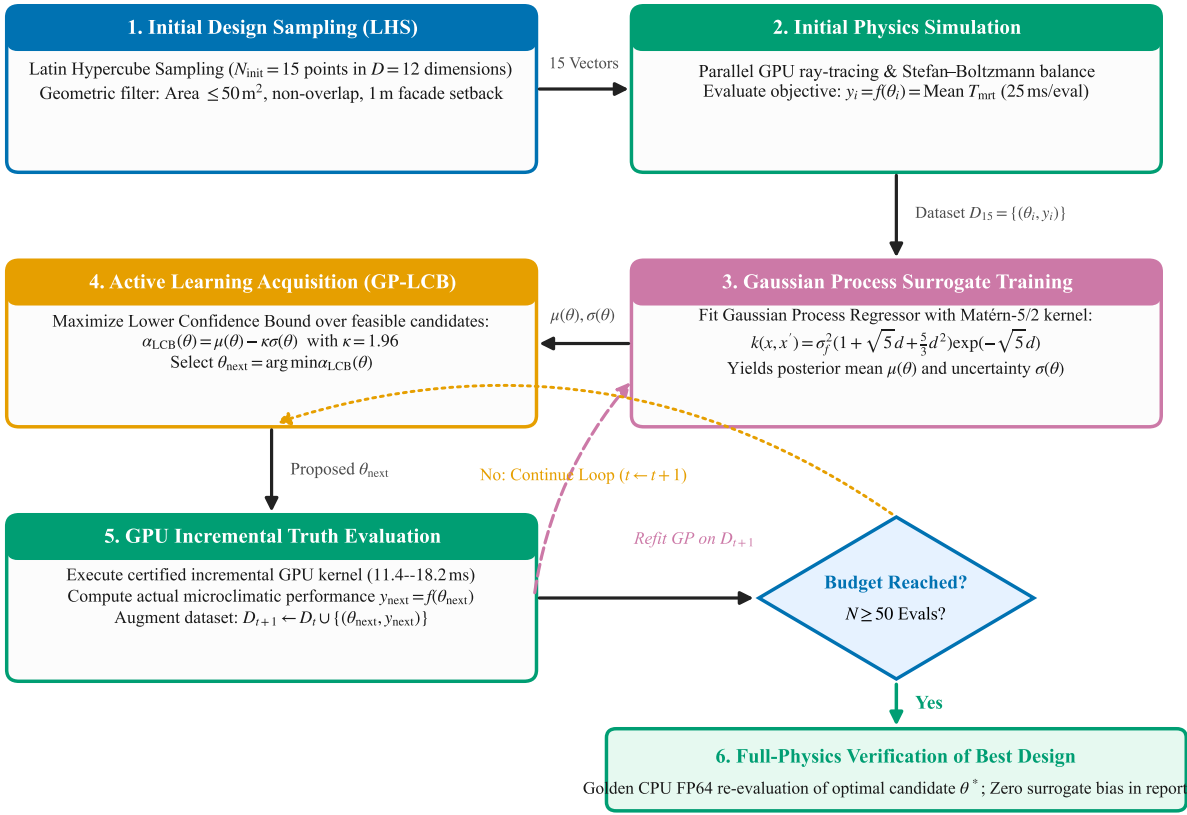

1. Initial Design Sampling (LHS)
Latin Hypercube Sampling ($N_{init} = 15$ points in $D = 12$ dimensions)
Geometric filter: Area $\leq 50\,m^2$, non-overlap, 1 m facade setback
15 Vectors
2. Initial Physics Simulation
Parallel GPU ray-tracing & Stefan–Boltzmann balance
Evaluate objective: $y_i = f(\theta_i) =$ Mean T_{mrt} (25 ms/eval)
Dataset $D_{15} = \{(\theta_i, y_i)\}$
4. Active Learning Acquisition (GP-LCB)
Maximize Lower Confidence Bound over feasible candidates:
$\alpha_{LCB}(\theta) = \mu(\theta) - \kappa\sigma(\theta)$ with $\kappa = 1.96$
Select $\theta_{next} = \arg\min \alpha_{LCB}(\theta)$
$\mu(\theta), \sigma(\theta)$
3. Gaussian Process Surrogate Training
Fit Gaussian Process Regressor with Matérn-5/2 kernel:
$k(x, x') = \sigma_f^2(1 + \sqrt{5}d + \frac{5}{3}d^2)\exp(-\sqrt{5}d)$
Yields posterior mean $\mu(\theta)$ and uncertainty $\sigma(\theta)$
Proposed θ_{next}
No: Continue Loop ($t \leftarrow t + 1$)
Refit GP on D_{t+1}
5. GPU Incremental Truth Evaluation
Execute certified incremental GPU kernel (11.4--18.2 ms)
Compute actual microclimatic performance $y_{next} = f(\theta_{next})$
Augment dataset: $D_{t+1} \leftarrow D_t \cup \{(\theta_{next}, y_{next})\}$
Budget Reached?
$N \geq 50$ Evals?
Yes
6. Full-Physics Verification of Best Design
Golden CPU FP64 re-evaluation of optimal candidate θ^*; Zero surrogate bias in reporting